Creating a DevOps Course

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ABSTRACT

Software engineering is regularly listed as a top career, but typical curricula rarely equip college graduates to join latest industry practices. Providing a course on software deployment and operations (DevOps) gives students a place to learn how the practices of agile software engineering align with organizational culture and business objectives. Students can get hands-on experience with current industry practices, tools, and technologies along with broader context of their utility and purposes. This work invites discussion on the structure of a recently developed course on DevOps, including course modules, assignments, discussions, and projects. Reflections from a pilot offering are given, along with lessons learned and planned changes for the next offering.

1 INTRODUCTION

US News ranked Software Developer as the #1 best job in their 2020 survey[1], and the US Bureau of Labor Statistics currently projects a 26% increase in Software Developer jobs by 2031[2]. Since typical programs in computing sciences require only one course on engineering software, graduates emerge with only basic ideas of the software development life cycle. Furthermore, there is rarely time in these courses to talk about the software maintenance phase. With the proliferation of the software-as-a-service (SaaS) model and the increasing reliance on cloud providers for deployment, students would benefit from a course covering software deployment and operations. In the context of a computing degree program, such a course would be able to provide technical, hands-on experience using current industry tools and practices. Moreover, such experiences could be placed into their motivating context of promoting organizational culture and achieving business objectives. With these motivations in mind, a new DevOps course was designed and piloted in summer of 2022. This work presents the course structure, instructional modules, assignments, lessons learned, and plans for future offerings. The author is happy to share course materials and collaborate on their improvement.

2 COURSE STRUCTURE

The following modules were produced after analyzing desired skills from relevant job ads.

(1) Review of agile software engineering (if needed)
(2) Agile business culture changes
(3) DevOps concepts and terms (CI/CD, etc.)
(4) Organizational culture and mindset required for DevOps
(5) Basics of cloud computing and connection to agile/DevOps
(6) Infrastructure as Code (IaC) and as a Service (IaaS)
(7) Building a DevOps pipeline from scratch
(8) From DevOps to DevSecOps

Module 1 is optional if the course follows one covering agile software engineering fundamentals. Modules 2-4 do not require a technical background but provide valuable context for students in computing sciences. Modules 5-8 ground the big ideas and lead to applied homework and projects.

3 ASSIGNMENTS AND PROJECTS

Choosing homework and projects of an appropriate scope is challenging. If too trivial, projects have no meaningful build and deploy stages; if too large in scope, novice students (and professors) are quickly overwhelmed by details peripheral to the learning outcomes. With the quickly changing landscape, choosing which DevOps tools and technologies to use in a course is another challenge. While utilizing the latest industry trends is good, the primary skill to impart is the ability to learn and adapt to new technologies quickly. The following assignments were required: Three homework assignments: Getting started with Docker, Basics of GitLab, Improve an existing GitLab pipeline; Four Discussions: Agile, Continuous Deployment, Docker Compose v. Kubernetes, DevOps Jobs (DevOps v. Site Reliability Engineers); Project options: New pipeline, Audit a pipeline, or Convert to GitHub Actions; Quizzes and final exam.

4 LESSONS LEARNED AND FUTURE PLANS

Students reported positive experiences but the compressed summer term was not ideal for the inevitable technical challenges of learning new tools and technologies. A typical semester would offer needed breathing room. The use of GitLab (free for educational use) offered valuable exposure to a popular industry tool, but students were at times overwhelmed with the learning curve; balancing this with a project option to convert a DevOps pipeline to use GitHub Actions was the most popular choice. Student discussions on technology trade-offs and career differences allowed students to paint in the context for their learning.

REFERENCES

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